Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

1. Batch Folder Completeness				
1.1	Batch folder exists with correct naming	Pass	Drive folder 'W4' (1Ed6zGEgZO69vZPbvmDo0l4Sw4zzZiZl0, owner garrett.lutz@, created 2026-09-27) holds 'INJ-09-2026-W4 Ops Notes', Scale Tickets/, Testing/, Kerry Oil/ (a separate Kerry removal) and the image 'RailLogix (Carville LA) to Basco B6 Truck Route - 136 mi.png'. Ops 'AECN Rail'!B2 reads INJ-09-2026-W4; Certify period 2026-09-19 to 2026-09-25. W4 has no 'Completed QA Checklists' subfolder yet. [log:drive_folder.root_items, drive_folder.folder, drive_folder.completed_qa_checklists_folder, ops_notes.batch_id_cell, certify.removal_period]	VR App. 1
1.2	Scale Tickets subfolder present and populated	Pass	Scale Tickets/ has one folder per railcar (SHPX215053, SHPX215039, SHPX215091 with offloads 1-4; SHPX215079 with offloads 3-4) and a Full and an Empty CAT ticket PDF in each offload folder: 28 files, all opened and read from the page image. Certify Sources carries the same 28 files, byte-identical (sha256, 28 of 28). [log:drive_folder.scale_tickets_tree, drive_folder.scale_ticket_files, scale_tickets.certify_vs_drive]	VR App. 1
1.3	Testing subfolder present with required lab results	Pass	Testing/ holds 'SR26-05719 W4 Kerry AECN QOWV COA.pdf'; page 2 is this removal's AECN certificate SR26-05719.002 (CLIENT ID 9-2026-W4-AECN); pages 1 and 3 belong to the Kerry and QOWV removals. [log:drive_folder.testing, lab_results.certificate, lab_results.client_id, lab_results.other_pages]	VR App. 1
1.4	Ops Notes spreadsheet present and complete	Flag	Google Sheet 'INJ-09-2026-W4 Ops Notes' (modified 2026-10-07T07:26:58Z; the 17 cells changed that morning sit on the Kerry tab and the Summary's Kerry rows, none on AECN) exported; every cell and comment on all 10 tabs read. The AECN Rail tab holds 4 railcars (rows 10-13), 9 BOLs (rows 17-25), 14 offload rows (28-41), the rollup (row 46) and a new railyard split (rows 48-50: RailLogix/Baton Rouge 77228.57392 kg, Opelousas 190578.5402 kg); the AECN Truck tab holds no deliveries. Gaps: (1) Section 2 lists 9 of the 12 BOLs: SHPX215079's loaded mass E13 77,870 kg is typed in and its BOLs 2453-R/2454-R/2455-R sit in W2's sheet. That follows the September guidance as fetched on 2026-09-29 ('Where a railcar is offloaded across more than one reporting week, the loaded mass is recorded once, in the week the railcar's first offload occurs, and subsequent weeks draw against the same shipment.'), and the older Ops copy in Certify explained it in M13, but the guidance as fetched 2026-10-07 no longer holds that clause and the live sheet has no M13 note, so the written basis survives only in superseded versions (8.3). (2) The AECN Rail tab carries no cell notes: the wellhead-pH note sits on AECN Truck H28 and Kerry D18, and the G46 note in the 2026-10-01 Certify copy is gone from the live sheet. (3) Row-9 headers still read 'Arrival Date at Opelousas' and 'Opel→Basco (mi)' over SHPX215053's Baton Rouge figures. (4) The two Ops Notes copies attached in Certify are earlier versions of this sheet (9.10). (5) The AECN Truck tab header reads 'BATCH HEADER — AECN Truck (Direct from Fort Lupton)', while every AECN BOL ships from Port-Cartier, Québec, and its empty table still carries N10:N12 2197 and N25 6591. P13 'PARTIAL' with Q13 'Y' is correct: P9 asks 'Full Railcar This Week?' and Q9 'Car Completes This Week?'. [log:ops_notes._primary_source_opened, ops_notes.cell_comments_check, ops_notes.cell_comments, ops_notes.railcars, ops_notes.railcar_headers_row9, ops_notes.railyard_split, ops_notes.SHPX215079_note, ops_notes.aecn_truck_tab, ops_notes.aecn_truck_tab.A1_header, bols.per_bol, vvb_guidance.removed_since_2026_09_29, ops_notes.change_since_first_read, certify.ops_notes_copies]	VR App. 1
1.5	Injection site confirmed as Basco 6, no Vaulted references	Flag	All 12 BOLs ship to CHARM Industrial Inc./WATCO, SOPOR Baie-Comeau; the two Basco route maps end at LA-115, St Landry LA 71367. Searching 1521 text fields (every Ops Notes cell on all 10 tabs, the 26 Certify component, datapoint and source names, the 12 BOL ship-to/ship-from/carrier fields, the 14 COBB rows) for 'Vaulted', 'Kansas', 'El Dorado', ', KS' and 'Purity Springs' gave 0 hits. One legacy site reference remains on the page images: page 5 of every one of the 12 BOL packages is Charm Purchase Order Q42025 dated 10/2/2025 (line 'Q32025', 6,000 MT RFO) with Ship To 'Charm Industrial, 838 SW Purity Springs Road, El Dorado, KS 67042', Charm's Kansas site. [log:vaulted_search, bols.per_bol, bols.po_page, route_evidence]	verify BOLs/Ops Notes/Certify
2. Scale Tickets & Mass Determination				

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

2.1	Empty and full scale ticket images present for each rail offload	Pass	Each of the 14 offloads has a Full and an Empty CAT ticket in Drive and in Certify Sources (28 of 28 byte-identical); all 28 read from the page image: ticket number, date, time, scale, axle weights, gross, tractor, trailer, commodity. All 28 are linked to 'Bio-Oil Mass - AECN via Rail'; none is linked to 'Mass of Injectant (per injection)' (9.10). [log:scale_tickets.offloads, scale_tickets.certify_vs_drive, certify.attachment_map, certify.attachment_gate]	VR Sec 5.1
2.2	Ticket dates within 2 weeks of injection	Fail	COBB and Ops date every offload's injection 2026-09-25. SHPX215079 breaks the rule 'For each pair: dates within 2 weeks of injection': offload 3 (full 1124426248529, 2026-09-05 09:29) and offload 4 (full 1124426248533 2026-09-05 14:20, empty 1124426249537 2026-09-06 07:50) were weighed 20 and 19 days before injection. They are the right railcar and no other week books them, but W2's offload 2 of the same railcar (full 1124426248528, 2026-09-05 09:24) was weighed five minutes before offload 3 at the same scale and booked in W2, while offloads 3-4 waited for W4; no cell note or guidance text says where their 35,459.148 kg was held (8.3). The other three railcars sit inside 14 days: SHPX215091 weighed 2026-09-17 11:22 to 2026-09-18 07:02, SHPX215039 2026-09-18 13:16 to 2026-09-19 16:17, SHPX215053 2026-09-22 16:16 to 2026-09-24 05:09. SHPX215091's four full weighs (2026-09-17) and SHPX215039 offloads 1-2 (2026-09-18) come before the Certify period opens on 2026-09-19: W3's Ops week (2026-09-11 to 2026-09-18) and W4's (2026-09-18 to 2026-09-25) both contain 2026-09-18, W3 did not book them, and COBB books every W4 offload on injection completion 2026-09-25, so nothing is counted twice, but no note states the booking basis. Each empty weigh follows its full weigh. [log:scale_tickets.offloads, scale_tickets.date_range, cobb.injection_completion_date, cobb.w4_order_line_completed_on, prior_weeks.w2_SHPX215079, prior_weeks.w3_ops_week, vvb_guidance.no_SHPX215079_timing_note, ops_notes.cell_comments_check, policy.status_rules, recompute.findings, _derived_values.ticket_day_gap_calc]	VR Sec 3.5
2.3	Tractor and trailer match between empty and full tickets, each offload	Pass	All 14 pairs match on tractor and trailer: CH1 with KL720, CH2 with KL188, and CH1 with KL192 (SHPX215079 offload 3); some tickets print the trailer with an 'RL' suffix ('KL720 RL', 'KL188 RL') on one ticket of the pair and without it on the other, the same trailer number. No standard empty weight is used this week. [log:scale_tickets.pair_comparison, scale_tickets.pairs_all_match, scale_tickets.label_notes, vvb_guidance.no_tare_substitution_section]	VR Sec 5.1
2.4	Net mass correctly calculated in Ops Notes (per offload)	Pass	Each offload's net recomputed from its two ticket images as (full lb - empty lb) x 0.453592 equals Ops F28:F41 to five decimals on all 14 rows (mass-chain scale_ticket_subtraction 14 of 14 PASS; recompute section 1), e.g. SHPX215053 offload 1: (73,800 - 29,400) lb gives 20,139.4848 kg. Ops D/E equal the 28 ticket gross weights. The 14 nets sum to Ops A46 268199.8778 kg. [log:mass_chain.counts, recompute.lines, ops_notes.offload_table, ops_notes.rollup.sum_deliveries_raw_kg, scale_tickets.offloads]	GHG Stmt Sec C
2.5	Injectate mass uses the conservative value (rail: scale tickets; loaded mass never over-drawn)	Pass	SHPX215053 (77,228.57392 kg offloaded against 77,570 kg loaded), SHPX215039 (77,573.30384 against 77,640) and SHPX215091 (77,546.08832 against 77,770) are fully drawn below their BOL totals, so their raw ticket nets stand. SHPX215079 closes here: W2 booked offloads 1-2 (21,799.63152 + 20,611.22048 kg, Ops H13 42,410.852), and offloads 3-4 weigh 21,137.3872 + 14,714.52448 kg, a total of 78,262.76368 kg against 77,870 kg loaded (BOLs 2453-R/2454-R/2455-R). The 392.76368 kg over-draw is cut from the last offload through the Uncertainty Adjustment J41, so offload 4 carries 14,321.7608 kg (77,870 - 63,548.24) and the railcar's loss is zero, as the last-load rule and the September guidance require. The mass-chain check confirms the cut (last_offload_derivation PASS, back-calculation triggered); its one FLAG is this cut. [log:mass_chain.last_offload, mass_chain.non_pass, mass_chain.fc11_flag_explained, policy.last_load_rule, vvb_guidance.quote_loss_method, ops_notes.railcars, ops_notes.offload_table, prior_weeks.w2_SHPX215079, recompute.lines]	GHG Stmt Sec C
2.6	Sparging deduction (rate 0.0129) applied correctly	Pass	Ops D46 3454.711772 kg is 0.0129 of the 14 delivered masses (constant O5 0.0129; per row in H28:H41, offload 4 of SHPX215079 on its cut mass), and recompute section 3 reproduces it. [log:ops_notes.rollup.sparging_kg, ops_notes.rollup.mass_used_kg, ops_notes.constants, recompute.lines]	GHG Stmt Sec C

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2.6a	Filtration loss deduction applied correctly (weekly methodology)	Pass	Ops E46 518.1307085 kg is 0.00196 x (delivered mass - sparging) summed over the 14 rows. Constant O7 0.00196 hyperlinks to 'AECN COA Solids Content Dataset (May 2025 - May 2026)' (title as Drive returns it), opened this session: Summary A12 'Recommended value for filtration loss', B12 0.001959191176, C12 'per-load mean — best represents what is actually delivered', over 136 loads and 71 certificates (B2 window '2025-05-18 to 2026-05-18'); the 'Materiality (Sec 2)' tab reads 'Proposed (fixed 0.196%)' (C12). O7's 0.00196 is B12 rounded up, which takes slightly more oil off the claim. [log:ops_notes.rollup.filtration_kg, ops_notes.constants, filtration_constant_source.summary_A12_B12_C12, filtration_constant_source.materiality_C12, filtration_constant_source.title_source, recompute.lines]	Weekly methodology
2.7	All mass values consistent across scale tickets/BOLs, Ops Notes and Certify	Fail	Consistent: Certify 'Mass of Injectant (per injection)' 263.8342716 t equals Ops F46 MOBI 263834.2716 kg; 'Bio-Oil Mass - AECN via Rail' 267.80711 t equals Ops C46; the Opelousas rail 'Mass of load' 190.57854 t equals Ops C50 and both Baton Rouge 'Mass of load' inputs 77.22857 t equal Ops C49. Not consistent: 'Opelousas to Basco - Tanker Truck Transport Process Emissions' takes its mass from 'Bio-Oil Mass - AECN via Rail' (267.80711 t, all four railcars) rather than the Opelousas mass 190.57854 t, so SHPX215053's 77.22857 t is charged on both the Opelousas truck leg and its own Baton Rouge truck leg. The guidance states 'the Opelousas distances are not applied to SHPX215053'. The leg is 1.313184 t against 0.934496 t on the right mass, 0.378688 t overcharged (conservative). While the BCU equals Certify's own process sum (9.6) this overcharge is offset one-for-one by the BCU and does not move the net, but the input is still wrong. Two further gaps: the Baton Rouge rail mass carries no +/- (7a.1), and the injectant mass and the three 'Mass of load' datapoints have no evidence linked in Certify (9.10). [log:certify.components, ops_notes.rollup, ops_notes.railyard_split, vvb_guidance.quote_br_method, recompute.findings, certify.attachment_map]	VR Sec 5.1
2.8	*Direct trucks: BOL-as-mass methodology*	*N/A*	*No direct trucks this week: the AECN Truck tab rollup A28 is 0 and rows B10:B24 are blank. [log:ops_notes.aecn_truck_tab]*	
3. Bills of Lading & Transportation				
3.1	Truck BOL(s) present for each shipment	Fail	Twelve inbound truck BOLs, one per AECN-to-SOPOR shipment, all opened and read page by page: 2497-R, 2498-R, 2500-R (SHPX215053); 2459-R, 2460-R, 2461-R (SHPX215039); 2465-R, 2466-R, 2467-R (SHPX215091); 2453-R, 2454-R, 2455-R (SHPX215079). The last three are in Certify Sources. The first nine are separate Drive files reached only through the hyperlinks in Ops 'AECN Rail'!B17:B25: they are not in the W4 batch folder and not among the 72 Certify sources. The knowledge base rule: '(1) check the batch Drive folder first, (2) if not found there, check Certify Sources tab, (3) only FLAG/FAIL if the document is absent from BOTH locations.' They are absent from both, so a verifier cannot confirm the loaded masses behind the 0.145 t loss and the three railcar balances (9.10). [log:bols.per_bol, bols.per_railcar, bols._how_obtained, bols.not_in_certify_sources, drive_folder, certify.presence_checks.w4_new_bols, policy.kb_document_location_rule_text, certify.attachment_map]	VR App. 1
3.2	Rail: 3 BOLs per railcar, destination SOPOR railyard	Pass	Three BOLs per railcar, each shipping to 'CHARM Industrial Inc. /WATCO, Société du port ferroviaire de Baie-Comeau, 18 Route Maritime, Baie-Comeau, Québec G4Z 2L6'; ship dates 2026-07-30 to 2026-08-31. [log:bols.per_bol, bols.per_railcar]	VR App. 1
3.3	BOL mass matches or exceeds reported injectate mass	Pass	Railcar level, on BOL 'Qty shipped': SHPX215053 77.57 MT, SHPX215039 77.64 MT and SHPX215091 77.77 MT each exceed their offloaded totals (77,228.57392, 77,573.30384 and 77,546.08832 kg); SHPX215079's 77.87 MT is matched exactly once offload 4 is cut (2.5). Every BOL ties internally: full - tare gives the delivered kg, delivered kg / 1,000 gives Qty shipped, the weighbridge 'Masse nette' equals the delivered kg, and litres at 60°F x density at 60°F lands within 0.59 kg of the delivered kg on all 12 (rounding allowance about 1.08 kg). [log:bols.internal_ties, bols.all_ties_hold, bols.per_railcar, ops_notes.railcars, mass_chain.railcar_balance]	VR Sec 3.5

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3.4	Origin and destination correct on all BOLs	Flag	All 12 BOLs ship from 'BioEnergie AE Côte-Nord Canada, 175 Portage des Mousses, Port-Cartier, Québec G5B 2V9' to SOPOR Baie-Comeau, carrier Transport Baie-Comeau; the rail leg is Watco (COBB Loads tab). Two naming gaps on the new leg: the guidance, the Certify components and Ops M10 call the transload 'Baton Rouge', while the only route evidence names 'RailLogix (Carville LA)' with origin 6045 Point Clair Rd, Carville, LA 70721; and the Basco end of that map is 4675 LA-115, St Landry, while the Opelousas map ends at 4687 LA-115. [log:bols.per_bol, cobb.loads_tab, vvb_guidance.quote_br_change, route_evidence, certify.certify_sources_enumeration]	PDD Sec A
3.5	Transportation distances consistent with known AECN routes and evidenced	Fail	(1) Baton Rouge rail: Certify 'Distance traveled' reads 2,153 km with no source. Ops K10 holds 2152 under 'Rail (mi)', the guidance gives 'Rail distance from the loading origin to Baton Rouge: 2,152 (map evidence is added in drive)', and the September GHG Statement Report v1.7 states 'Its rail leg (2,152 mi)'. 2,153 km is 1,337.8 mi, about 61 % of Watco's 2,199 mi to Opelousas. On 2,152 mi the leg is 4.306208 t against Certify's 2.676997 t, understated by 1.629211 t. Four Drive searches on 2026-10-07 (titles; full text '2,152' and '2152 mi'; and Certify's own figure '2153' or '2,153', whose 50 results hold it only as BOL '2153-T' and in no distance), the W4 folder, the Rail Logix Lessons Learned document and the 72 Certify sources hold no SOPOR-Baton Rouge rail distance record. (2) Baton Rouge truck: 136 mi agrees with the guidance and is the longest of the three routes on the map in the W4 Drive root (121 / 117 / 136 mi), as the statement report's longest-route rule requires, but the map is not linked in Certify. (3) The rail allocated distance is round trip this week: Ops M42 15,102.68824 mi (2 x 2,152 + 2 x 2,199 + 2 x 2,199 + 2 x 2,199 x SHPX215079's share) against one way in W1-W3 and in the 2026-10-01 Ops copy (7,551.34412 mi), so W4 charges 0.235078 t more than the September weeks before it; the guidance names the round trip for the tanker leg only. The railcar factor is 'Embodied Emissions per Mile Traveled (MTCO2e)' with no empty-return uplift, so the round trip fits it and the one-way basis of W1-W3 charged fewer miles than the railcars travelled; the basis needs settling for the period, not W4 reverting to one way. SHPX215079 now carries 3,200.34412 mi across W2 (one way) and W4 (round trip), neither basis for the railcar. (4) 'Allocated Distance AECN to Sopor Quebec (truck)' 1,111.70083 mi follows the Ops method on 107.49722 mi, not the attached-sheet L42 1,111.729637 (effect -0.0000068 t); it is linked to four Mangrove maps of the AECN-to-Opelousas road route (3,672.1 km) and not to the Ops Notes that holds the allocation. (5) 'Allocated Distance Opelousas to B6 (truck)' 1,946 mi (42.9 x 2 x 10 + 136 x 2 x 4) has no source. (6) The Watco email names no railcar, route or date. AECN-SOPOR 107.49722 mi (173 km map) and Opelousas-Basco 42.9 mi reproduce. [log:certify.components, certify.modal_reads, ops_notes.railcars, ops_notes.rollup, vvb_guidance.quote_br_values, vvb_guidance.quote_br_method, rp_status.statement_report_v1_7, route_evidence, certify.presence_checks.baton_rouge_route_evidence, certify.sources_downloaded_and_opened, certify.ops_notes_copies, prior_weeks.w2_SHPX215079, certify.attachment_map, route_evidence.search_2153, standard_efs.railcar_embodied_build, recompute.findings]	Transport Emissions Acctg v1.1
3.6	Transport mode matches oil type and route	Pass	Eight transport components: AECN-to-SOPOR tanker truck (process + embodied), SOPOR-to-Opelousas rail (process) and SOPOR-to-Baton Rouge rail (process) with one shared rail embodied component, Opelousas-to-Basco and Baton Rouge-to-Basco tanker truck (process) with one shared truck embodied component. [log:certify.components]	Certify Datapoints
3.7	Billing documents present and consistent with BOL	Flag	Each BOL package bundles the Straight BOL, Shipping Details, the BioEnergie weigh ticket, the TSCA certificate and a Charm PO; all 12 read, and each weigh-ticket 'Masse nette' equals the BOL delivered mass. Hygiene: the page-5 PO is Q42025 of 10/2/2025 (line 'Q32025', Ship To El Dorado, KS; 1.5) with the seller signature blank on 8 of the 12 (2497-R, 2498-R, 2500-R, 2461-R, 2465-R, 2466-R, 2467-R, 2453-R); 2466-R is a corrected package ('***Correction of density and final liters', density 1.1980 and litres 21,586 hand-corrected; masses unchanged); 2498-R prints CofA 'RS2-260825-1' on page 1 and 'RS2-260828-1' on Shipping Details; 2453-R prints production period '#0X-2026'. [log:bols.per_bol, bols.po_page, bols.hygiene]	VR App. 1
4. Bio-Oil Composition & Testing				

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

4.1	CHN analysis PDF present	Pass	SGS Certificate of Analysis SR26-05719.002 (SGS order 5506143), page 2 of the three-page report in Drive Testing/, read this session. It is not linked to 'Carbon Content of Bio-oil Injected (per injection)' in Certify (9.10). [log:lab_results, drive_folder.testing, certify.presence_checks.sgs_certificate, certify.attachment_gate]	VR App. 1
4.2	Carbon content within expected AECN range	Pass	Carbon 44.0 % (m/m), marked '#' ('Result is outside of test method limits and/or analytical range used in method precision study'). MRV_AECN_Analysis.xlsx Summary Statistics (opened this session): mean 39.420350877193 %, SD 6.09332409018271 %, n 285, so the two-SD band is 27.23 % to 51.61 %; the batch sits 0.752 SD above the mean. Ops Summary E17 0.44 and Certify Carbon Content 44 % match. W3 read 39.7 %. [log:lab_results.carbon_pct, lab_results.hash_footnote, lab_results.w3_comparison, carbon_range, ops_notes.summary_aecn, certify.components, _derived_values.carbon_z_calc]	Biomass Feedstock Acctg v1.2
4.3	Hydrogen and nitrogen values reported and reasonable	Pass	Hydrogen 6.6 % (marked '#'; dataset mean 6.87438596491228 %, SD 0.562848694875852 %); Nitrogen '<0.75' % (m/m), below the reporting limit. [log:lab_results.hydrogen_pct, lab_results.nitrogen_pct, carbon_range]	Biomass Feedstock Acctg v1.2
4.4	Lab report references correct batch/sample ID	Flag	CLIENT ID '9-2026-W4-AECN', SAMPLED 25-Sep-2026 14:00 (Ops Summary D17 2026-09-25, C17 'SR26-05719.002'), RECEIVED 28-Sep-2026, report 28-Sep-2026: the identity ties. The LOCATION field prints 'FORT LUPTON, CO' on the AECN page while the AECN composite is drawn at Basco, Louisiana; W3's certificate printed the same location. [log:lab_results.client_id, lab_results.sampled, lab_results.received, lab_results.location_on_certificate, ops_notes.summary_aecn]	VR Sec 5.1
4.5	*pH/conductivity (aqueous phase)*	*N/A*	*AECN is not aqueous phase. [log:oil_type]*	
4.6	Density above formation fluid threshold for safe injection	Pass	Density at 15°C 1.203 g/cm³. Formation fluid: Waypoint Analytical report 24-200-0020, Lab No 98021, page 5, 'Specific Gravity 1.01 g/mL', ASTM D-1298-85 (opened this session). The bio-oil is denser than the formation fluid. [log:lab_results.density_15C_g_cm3, lab_results.formation_fluid]	UIC Permit / safe injection
5. Ops Notes Accuracy & Internal Consistency				
5.1	Correct batch ID and injection week in Ops Notes	Flag	'AECN Rail'!B2 reads INJ-09-2026-W4, week 2026-09-18 to 2026-09-25; all 14 offload rows dated 2026-09-25; Certify period 2026-09-19 to 2026-09-25, inside the guidance's W4 window '18 – 25 September 2026'. Label error carried from W3: A3/A4 read 'Week Start (Sat)' and 'Week End (Sat)' but 2026-09-18 and 2026-09-25 are Fridays, and 2026-09-18 is also W3's end date (no offload is booked in both weeks). [log:ops_notes.batch_id_cell, ops_notes.week_start, ops_notes.week_end, ops_notes.week_labels, ops_notes.offload_table, certify.removal_period, vvb_guidance.quote_br_scope]	verify header
5.2	Scale ticket values correctly transcribed into Ops Notes	Flag	Ops D/E equal the 28 ticket gross weights exactly (recompute section 1). Ticket labels are inconsistent: both SHPX215039 offload 2 tickets (1124426261598, 1312226261663) carry commodity 'SHPX215039 O1', the label of offload 1; SHPX215053 offload 4's empty ticket 1186426267016 reads 'SHPX215053' with no offload number; SHPX215053 labels vary ('SHPX215053 01', 'SHPX 215053 OFF 2', 'SHPX215053 O3'); the four SHPX215079 tickets print no delivery site. [log:scale_tickets.label_notes, scale_tickets.offloads, recompute.lines]	VR Sec 5.1
5.3	*Oil and aqueous phase mass reconciliation*	*N/A*	*AECN-only removal, no aqueous phase. [log:oil_type]*	
5.4	COBB values present and match Ops Notes/Certify	Pass	COBB tab 'Basco Injection – COBB' rows 3724-3737 (latest import 2026-10-05 12:24:22), column A 'INJ-09-2026-W4_bio_oil' confirmed before reading: all 14 offloaded_quantity_kg equal the Ops delivered masses (mass-chain cobb_offload_consistency 14 of 14 PASS), offload 4 of SHPX215079 at the cut 14,321.76 kg; origin_quantity_kg equals each railcar's BOL total (77,570 / 77,640 / 77,770 / 77,870); BOL_number lists each railcar's three BOLs; injection_completion_date 2026-09-25. [log:cobb.tab, cobb.row_id_at_qa, cobb.batch_id_col_a, cobb.offloaded_quantity_kg, cobb.origin_quantity_kg, mass_chain.counts]	COBB tracker
5.5	pH measurement noted	Pass	pH 2.97 on all 14 offload rows (G28:G41) and in the rollup G46, inside the O6 target '2.0 - 4.0'. The method is recorded as a cell note on AECN Truck H28 and Kerry D18: 'Wellhead pH = weekly median of Ignition PH-1041, gated to StateRun=1; provided by G. Lutz 2026-09-30.' pH is not a Certify datapoint. [log:ops_notes.offload_table, ops_notes.rollup.pH, ops_notes.constants, ops_notes.cell_comments]	GHG Stmt Sec C
5.6	Completion date present and matches COBB	Pass	COBB injection_completion_date 2026-09-25 on all 14 rows; Ops offload rows dated 2026-09-25; Certify period ends 2026-09-25. [log:cobb.injection_completion_date, ops_notes.offload_table, certify.removal_period]	VR Sec 3.5

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

5.7	*No erroneous density measurements*	*N/A*	*Density field expected blank for AECN in Ops Notes; the lab density is used only for the safe-injection check (4.6). [log:oil_type]*	
6. AECN Feedstock Emissions & Allocation				
6.1	AECN process emissions LCA CI spreadsheet present	Fail	The vendor workbook 'Bioenergy AE Cote-Nord Canada - LCA CI calculation - 1Jan2026 to 30Jun2026.xlsx' is not in this removal's 72 Certify sources and not in the W4 Drive folder. The pyrolysis-process CI is linked to 'FH 2026 LCA w Stack Emissions.xlsx' only, whose I11 0.133052027800741 is labelled 'From Vendor LCA'; W3's removal carried the vendor workbook on the same datapoint, so the vendor half of 0.13712 has no backing on W4. [log:lca_files, certify.presence_checks.vendor_lca_files, drive_folder.root_items]	VR App. 1
6.2	LCA with stack emissions file present	Pass	'FH 2026 LCA w Stack Emissions.xlsx' is linked to the pyrolysis-process CI; Sheet1 I12 0.1371171258 (stack I10 0.004065098048 + vendor I11 0.133052027800741). [log:lca_files.fh_2026, certify.attachment_map]	VR App. 1
6.3	Supporting GHG data file present	Fail	'Supporting GHG data 30Jun2026 S.xlsx', linked on W3's removal, is not in this removal's 72 Certify sources and not in the W4 Drive folder. [log:lca_files, certify.presence_checks.vendor_lca_files, drive_folder.root_items]	VR App. 1
6.4	AECN pyrolysis process EF matches current half-year value	Pass	Certify CI 0.13712 tCO2e/t against Standard EF sheet 2026 tab row 4 'Production Emissions for AECN Oil--H2 2026', Final Quantity 0.1371171258 (source 'FH 2026 LCA w Stack Emissions'); the removal falls in H2 2026. Certify's value is the sheet value rounded up at the fifth decimal, 0.000769 t more emission (recompute section 5). [log:standard_efs.row4_aecn_process_H2_2026, standard_efs.period_basis, standard_efs._primary_source_opened, recompute.lines]	Certify Component Inputs
6.5	AECN pyrolysis embodied EF applied	Pass	Certify CI 0.02116 tCO2e/t; Standard EF sheet row 5 Final Quantity 0.02116; 'AECN Embodied Emissions Calc Sheet (1).pdf' (linked, opened) gives 0.02115941123. [log:standard_efs.row5_aecn_embodied, standard_efs._primary_source_opened, lca_files, certify.attachment_map]	Certify Datapoints
6.6	Mass basis for pyrolysis EFs is pre-sparging origin mass	Pass	Both pyrolysis components use 'Bio-Oil Mass - AECN via Rail' 267.80711 t, the 14 delivered masses before sparging (Ops C46, with SHPX215079 offload 4 at its cut mass), as the guidance sets ('calculated on the delivered mass: scale-ticket net mass, less any Uncertainty Adjustment'). The 632.03392 kg lost from SHPX215053, SHPX215039 and SHPX215091 is charged its pyrolysis emissions through the separate 'AECN Bio-oil Loss' component (8.2), so all oil that entered custody carries its upstream emissions. [log:certify.components, ops_notes.rollup.mass_used_kg, vvb_guidance.quote_loss_method, vvb_guidance.quote_loss_purpose, policy.custody_rule, ops_notes.railcars]	Certify Components
7. Gross-to-Net Calculation				
7.1	Gross CO2e = injectate mass x carbon wt% x 3.667	Pass	263.8342716 t x 0.44 x 3.667 gives 425.691321 t, Certify's Injection sequestration; the mass input equals Ops F46 MOBI and the carbon equals SGS SR26-05719.002 (recompute sections 3 and 5). No R&D removal this week (K46 blank). [log:certify.components, ops_notes.rollup.mobi_kg, ops_notes.rollup.mass_removed_rnd_K46, lab_results.carbon_pct, _derived_values.gross_calc, recompute.lines]	GHG Stmt Sec E
7.2	Process emissions correctly deducted	Fail	Activities 83.620 t in the Calculation view equal the AECN components 63.069291 t (pyrolysis 42.388509, transport process 19.339248, transport embodied 1.196534, loss 0.145) plus the site components 20.550448 t, and every component recomputes from its own inputs. Two AECN legs are wrong against the evidence: the Baton Rouge rail leg is understated by 1.629211 t (3.5) and the Opelousas truck leg overstated by 0.378688 t (2.7), so activities are understated by 1.250523 t net; on the evidence they are 84.869836 t (recompute CORRECTIONS D). The transport part of this error is offset one-for-one by the BCU while the BCU equals Certify's own process sum, so it does not move the net (9.8). [log:certify.calc_view, certify.components, certify.activities_site_t, _derived_values.activities_calc, recompute.findings]	GHG Stmt Sec E

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

7.3	Uncertainty discount correctly applied	Pass	Calculation view: 'Net CO_2e Removed 362.068 t' from Sequestrations 425.691, Adjustments 22.439, Activities 83.620 and Uncertainty Discount 2.443. Re-derived from the component inputs as one combined one-sigma, one term per datapoint: carbon +/- 0.176 % (1.702765 t), injectant mass +/- 1,015.98 kg (1.639263 t), grid electricity +/- 1,271.5038 kWh at the grid factor 0.4630140342 (0.588724 t), Bio-Oil Mass +/- 1,015.98 kg on the four components that share it (0.178274 t), the Opelousas rail and Baton Rouge truck masses, site diesel and gasoline; this reproduces 2.4428820922 t (recompute section 9) on the inputs as entered; the inputs that lack a +/- are carried in 7a.1, 0.57 % of gross. [log:certify.calc_view, certify.uncertainty_discount_t, recompute.lines, _derived_values.discount_pct_calc]	Isometric Standard v2.0 S3.7
7.4	Net removals consistent with BCU Quant and GHG Statement	Flag	Certify's GHG statements list (read today) ends at '01 Aug 2026 to 31 Aug 2026' (Certificates issued, 11 Sep 2026, 1,746.14 tCO_2e); the September statement is not in Certify yet. Two September drafts sit in Drive ('Additional Documentation', garrett.lutz@), both read this session. The results workbook 'September1-September30_Bio-Oil_GHG_Statement_Results.xlsx' (edited 2026-10-07T07:43:32Z, when the W3 and W4 Kerry rows swapped values; the W4 AECN row and the totals did not change) carries W4 AECN as 263,834.27 kg injected, gross 425.691, process 63.0690, site 20.5500, reductions 22.4390, discount 2.4429 and net 362.07 t, matching Certify to its rounding (recompute section 11). The 'GHG Statement Report v1.7' (dated 10/3/26, modified 2026-10-05T16:07:59Z) does not match that workbook: Section E process 384.65 against 385.33 t, site 20.48 against 20.55 t, reductions 171.95 against 172.66 t, net 1,744.01 against 1,743.96 t; its Section B gives W4 site emissions 20.482 t with Additional Transport 1.084 t, against Certify's 20.550448 t and 1.15242 t. Both drafts carry Certify's W4 AECN process figure 63.069 t, not the evidence figure 64.319388 t (7.2). [log:rp_status, recompute.findings, recompute.lines, certify.components]	RP BCU Quant.xlsx
7a. Per-Component Uncertainty Verification				
7a.1	Layer 1 - every component that needs uncertainty has it on the right input	Fail	All 13 AECN component windows opened. Injection sequestration shows +/- on carbon and mass; both pyrolysis components, the AECN-SOPOR and Opelousas truck legs, the Opelousas rail leg and the Baton Rouge truck leg show +/- on their mass input; the three embodied components are per-mile with no mass input; the BCU is a constant reduction. 'SOPOR to Baton Rouge Rail Process Emissions' shows 'Mass of load 77,228.57 kg' with no +/-, where the same mass on the Baton Rouge truck leg carries +/- 290.28 kg (4 x 72.57). The uncertainty reference requires it ('72.57 kg per discrete weight'), and leaving it out understates the discount: 2.4428820922 to 2.4429028145 t on today's inputs, 2.442586 to 2.442640 t once the distance and the Opelousas truck mass are corrected (recompute CORRECTIONS B to C). Missing required data is a Fail at any size ('This includes: reporting errors, data mismatches, missing required data, conflicting evidence'); the ~0.001 t allowance applies to matching values, not to a missing input. It belongs in the same edit as the 3.5 distance on that component. The 'AECN Bio-oil Loss' constant also carries none. On the same removal, outside this batch's scope (site-emissions QA): 'Argon Electricity Usage (RECs)' grid use 2,602.43355 kWh carries no +/- although the knowledge base lists 'Electricity → Grid Electricity Usage: 0.02 × Measured Value (2%)' (with it the discount rises by 0.0001189 t), and the procured-power input carries none on either site electricity component against 'Electricity → Procured Power Electricity Usage: 0.02 × Measured Value (2%)'. [log:certify.modal_reads, certify.components, policy.status_rules, policy.kb_electricity_uncertainty, recompute.findings]	MRV Uncertainty Cheat Sheet
7a.2	Layer 2 - reported +/- matches the expected value	Pass	Where present the +/- is right: 1,015.98 kg on the injectant mass and on 'Bio-Oil Mass - AECN via Rail' (14 offload weighings x 72.57 kg), 725.7 kg on the Opelousas rail mass (10 offloads), 290.28 kg on the Baton Rouge truck mass (4 offloads); carbon 0.176 % against 0.004 x 44.0. One outbound weighing per offload (Failure Category 9). [log:certify.modal_reads, certify.components, ops_notes.summary_aecn, ops_notes._primary_source_opened, _derived_values.unc_calc, recompute.lines]	Isometric Standard S3.7.3

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

7a.3	Layer 3 - discount present and consistent	Pass	Uncertainty Discount 2.4428820922 t is present in the Calculation view, carried into the net, and reproduces from the per-datapoint inputs (7.3; recompute section 9). Against the September drafts: the results workbook carries 2.4429 t on the W4 AECN row (M10), and its discount column M2:M12 sums to 10.9981 t against the GHG Statement Report v1.7 Section E 'Uncertainty Discounts' 10998.10 kg / 11.00 t; both agree to their rounding (recompute sections 11 and 12). The September statement is not in Certify yet (7.4), and the discount rests on inputs that still lack the Baton Rouge rail +/- (7a.1). [log:certify.uncertainty_discount_t, certify.calc_view, rp_status.results_workbook.discount_column_sum_t, rp_status.statement_report_v1_7.section_e_discount_lines, rp_status, certify.modal_reads, recompute.lines]	Isometric Standard S3.7.3
8. Data Integrity & Anomaly Check				
8.1	No unexplained outliers in mass, carbon or calculated values	Flag	Nets run 32,440 to 49,140 lb, full gross 60,700 to 78,620 lb, carbon 0.752 SD above the mean. Eight tickets read 0 on the steer axle, all on tractor CH1 with trailer KL720: SHPX215053 offload 4 empty, SHPX215039 offloads 2 and 3 (full and empty), SHPX215091 offload 2 (full and empty) and offload 4 full; every other ticket reads 9,340 to 12,480 lb on the steer. The CH1/KL720 empties weigh 29,240 to 29,640 lb whether or not the steer reads 0 (29,400 and 29,580 lb with a steer reading), and each gross is the sum of its readings, so the nets stand. The SGS acid number is 156 mg KOH/g (marked '#'), against 61 in W2 and 47 in W3; it is not a Certify input, and no note explains the threefold rise. The Kerry sample in the same SGS order 5506143 reads 154 mg KOH/g, also marked '#', and the QOWV sample 36, so the jump shows in two oils from one lab run. [log:scale_tickets.zero_steer_tickets, scale_tickets.zero_steer_rig, scale_tickets.ch1_kl720_empties_lb, lab_results.acid_number_by_week, lab_results.same_order_acid_numbers, lab_results.hash_footnote, _derived_values.ticket_ranges_calc, _derived_values.carbon_z_calc]	GHG Stmt Sec C
8.2	Spills, losses and irregularities documented	Fail	SHPX215053 lost 341.42608 kg (0.440 %), SHPX215039 66.69616 kg (0.086 %) and SHPX215091 223.91168 kg (0.288 %) between loading and offload, recorded in Ops G10:G12; SHPX215079's loss is zero after the cut (2.5). Their upstream emissions are charged through 'AECN Bio-oil Loss' 145 kgCO2e, as the guidance sets: 'Loss emissions apply the same AECN pyrolysis carbon intensities (process and embodied) and the same loading-origin-to-rail-yard transport emissions used for the delivered oil, pro-rated to the lost mass'. The evidence linked to that figure does not give it. (1) The linked Ops copy ('- 2026-10-01') shows the loss column G10:G13 as 341.42608, 66.69616, 223.91168 and -392.76368 kg (G13 '=IF(AND(E13<>"",F13<>"",F13>=0.95*E13),E13-F13,"")'), summing to 239.27024 kg, which at Certify's own rate per tonne lost gives 0.054891 t. 0.145 t needs 632.03392 kg, which drops G13; the live sheet does that through MAX(0, E13-F13), and the statement report v1.7 says 'Where scale tickets exceed Bills of Lading, the excess is removed through the Uncertainty Adjustment and no negative loss is applied.', but neither is attached. (2) The linked example image (SHPX215053, 341.43 kg, 0.07833 t, 'Opel to B6') puts SHPX215053 on the Opelousas distances, which the guidance excludes ('the Opelousas distances are not applied to SHPX215053'); 0.145 t reproduces (0.144996 t) on that basis only. On the guidance basis, SHPX215053 on its own 2,152 mi rail leg, the loss is 0.144574 t, so Certify charges 0.000426 t more (conservative). (3) The BOLs behind the loaded masses are not attached (9.10), and the constant carries no +/- (7a.1). The loss also charges rail embodied on the lost share that the rail allocation already gives each full railcar in whole: 0.000557 t counted twice (conservative). [log:certify.ops_notes_copies, certify.ops_notes_copies.loss_positive_sum_kg, rp_status.statement_report_v1_7, ops_notes.railcars, vvb_guidance.quote_loss_factor, vvb_guidance.quote_loss_evidence, vvb_guidance.quote_br_method, certify.sources_downloaded_and_opened.loss_example, certify.components, policy.custody_rule, policy.status_rules, recompute.findings, recompute.lines, _derived_values.loss_calc]	GHG Stmt Sec C

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

8.3	Inventory reconciliation / carryover documented	Fail	SHPX215079 spans W2 and W4: W2 booked offloads 1-2 (42,410.852 kg) and W4's H13 'Previously Claimed' carries that figure, so the railcar closes at its BOL total with the cut on offload 4 (2.5). All four offloads were weighed 2026-09-04 to 2026-09-06, offloads 1, 2 and 4 by the same CH2/KL188 rig, and W2's offload 2 (09:24) and W4's offload 3 (09:29) five minutes apart on 2026-09-05; offloads 3-4 were then injected in W4, three weeks later, and W3 did not book them. Nothing records where the 35,459.148 kg sat in between, so the carryover is not documented as the check requires ('If inventory reconciliation applies: carryover oil from prior production periods documented and allocated'). COBB books oil in the week it is injected: rows 3781-3792 (SHPX215061, SHPX215067 and SHPX215073, order lines completed 2026-09-22 to 2026-09-29) sit in INJ-10-2026-W1 with injection completion 2026-10-02, none of them in W4, and the Rail Logix notes record further RailLogix loads on 9/24 and 9/25 ('Starting on the 2nd load at 0915') without naming the railcar. The cross-week method rests on a guidance clause present in the 2026-09-29 version ('Where a railcar is offloaded across more than one reporting week, the loaded mass is recorded once, in the week the railcar's first offload occurs, and subsequent weeks draw against the same shipment.') and missing from the version fetched 2026-10-07; even that clause covers the loaded mass, not when the oil is injected (1.4). [log:ops_notes.railcars, prior_weeks, scale_tickets.offloads, cobb.late_september_railcars, route_evidence.rail_logix_lessons_learned, vvb_guidance.removed_since_2026_09_29, vvb_guidance.no_SHPX215079_timing_note, ops_notes.cell_comments_check, policy.status_rules, recompute.findings]	VR Sec 3.5
8.4	Rail uncertainty conservatism: offloads at least equal inbound BOLs	Pass	Each railcar drew down in 4 offloads against 3 inbound BOLs (SHPX215079: 2 in W2 and 2 here). Per-offload uncertainty (7a.2) books more than the inbound weighings would. [log:mass_chain.rail_uncertainty_precondition, bols.per_railcar]	Isometric Standard S3.7.3
9. Isometric Certify - Removal Component Mapping				
9.1	SEQUESTRATIONS: gross injection sequestration matches internal gross	Pass	Injection sequestration 425.691321 tCO2e matches the internal gross from Ops F46 MOBI and SGS carbon 44.0 % (7.1). [log:certify.components, _derived_values.gross_calc, recompute.lines]	Certify Components
9.2	ACTIVITIES - Transport: process + embodied correct for all legs	Fail	Recomputed from the Standard EF sheet 2026 tab (tanker 0.0001143 MT/(t·mi) and 0.0002375605888 MT/mi; rail 2.5910374e-05 MT/(t·mi) and 3.113066508e-05 MT/mi) on the evidence masses and distances. Correct: AECN-SOPOR truck 3.290528 t; Opelousas rail 10.858420 t (sheet 10.858577); Baton Rouge truck 1.200119 t (sheet 1.200503: Certify uses the GLEC library 'Tanker truck, Northern America, WTW' 0.071 kgCO2e/(km·t) against the sheet's 0.0710227273, 0.000384 t lower); embodied 0.264096 t, 0.470147 t and 0.462292 t. Wrong: (1) SOPOR-Baton Rouge rail 2.676997 t on '2,153 km'; on 2,152 mi it is 4.306208 t (4.306198 t on the sheet rail factor), 1.629211 t understated (3.5). (2) Opelousas-Basco truck 1.313184 t on all four railcars' mass; on the Opelousas mass it is 0.934496 t, 0.378688 t overstated (2.7). (3) The rail embodied leg is round trip this week (15,102.69 mi) against one way in W1-W3, 0.235078 t more than the September method before it (3.5). Both Baton Rouge legs cite the 'GLEC V3.2' library source rather than the Standard EF sheet used on every other leg. (4) The Opelousas/Baton Rouge to Basco tanker embodied leg counts the empty return twice: its 1,946 mi is already a round trip ('The tanker leg is counted as a round trip') and the sheet factor is 'Tanker Truck Total Embodied Emissions (MTCO2e)/Mile w/ Deadhead' (=B12*(1+B14), deadhead share 0.438); without the uplift the leg is 0.321483 t against Certify's 0.462292 t, 0.140809 t over (conservative; the earlier September weeks also use the round trip). Factor rounding: pyrolysis CI 0.13712 against 0.1371171258 (+0.000769 t). [log:standard_efs, standard_efs._primary_source_opened, certify.components, certify.sources_downloaded_and_opened.glec_library_source, standard_efs.tanker_embodied_build, vvb_guidance.quote_br_method, recompute.findings, recompute.lines, _derived_values.activities_calc]	Certify Components
9.3	*ACTIVITIES - Injection: LCS / sample transport*	*N/A*	*AECN needs no LCS pre-treatment; none of the 13 AECN components is an injection-side activity. [log:certify.components]*	*Certify Components*
9.4	ACTIVITIES - Pyrolysis: process + embodied entries present	Pass	'AECN pyrolysis (from rail loads) - process emissions' 36.721711 t (267.80711 t x 0.13712) and '... - embodied emissions' 5.666798 t (267.80711 t x 0.02116), both recomputed. [log:certify.components, recompute.lines]	Certify Components

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

9.5	REDUCTIONS: applicable BCU categories populated	Fail	'BCU - Transport Process', Constant CO_2 reduction 19.339 tCO_2e, has no source linked; the retirement certificate (Charm-RetirementCertificate-RD-EACs-20251218.pdf, 2,795 Renewable Diesel TERCs, 13-47206-50000) linked on W3's BCU is not among this removal's 72 sources. Transportation Emissions Accounting v1.1 §4.0 allows BCUs ('the use of Book and Claim Units (BCUs) for low-carbon fuels to substitute for some, or all, of project fuel usage is permitted'), and EC1 requires 'proof of BCU purchase and retirement in a volume equivalent to the volume of fuel substitution claimed.' Transport process is the only BCU category among the AECN components. [log:bcu_sources, certify.presence_checks.bcu_certificate, certify.attachment_map, certify.components]	Certify Components
9.6	REDUCTIONS - BCU does not exceed transport process emissions	Flag	BCU 19.339 t against the transport process sum 19.3392476 t (AECN-SOPOR 3.290528 + Opelousas rail 10.858420 + Opelousas truck 1.313184 + Baton Rouge rail 2.676997 + Baton Rouge truck 1.200119): it sits 0.0002476 t below, so it does not exceed, but it is Certify's own process sum rounded down by 0.2476 kg, which the exact-figure rule also excludes ('fix = enter the exact process figure, never round down either'). Equation 5 of §4.2.1 allows at most the BCUs for which '(F_j − $F_{BCU,j}$ $ED_{BCU,j}/ED_{Fuel,Transportation,j}$) ≥ 0'. The fixes in 3.5 and 2.7 move the sum to 20.5897707 t. They must land together: corrected alone, the Opelousas truck mass drops the sum to 18.9605596 t and the BCU would then exceed it by 0.3784404 t. Charm's rule is the exact process figure, so the BCU entry follows the corrected sum and draws 1.2507707 t more on certificate 13-47206-50000 (S.1). [log:bcu_sources, policy.bcu_cap_rule, policy.bcu_fix_rule, certify.components, recompute.findings, _derived_values.bcu_calc]	Certify Components / Calculation View
9.7	COUNTERFACTUALS: zero per protocol	Pass	The Calculation view carries no counterfactual term: Sequestrations, Adjustments, Activities and Uncertainty Discount only (7.3). [log:certify.calc_view]	Certify Components
9.8	Net removal matches internal calculation	Flag	362.068 t reproduces from Certify's own terms (recompute 362.068120). On the evidence, with the BCU set to the corrected transport process sum as Charm's exact-figure rule requires (9.6), the net is 362.069035 t, 0.001035 t above Certify (recompute CORRECTIONS E): the transport errors in 3.5 and 2.7 cancel against the BCU, which equals Certify's own understated process sum. The net holds only while those errors and the BCU move together: correcting the activities alone gives 360.818265 t (CORRECTIONS D), and correcting the Opelousas truck mass alone puts the BCU 0.3784404 t over its emission (9.6). The Baton Rouge distance, the Opelousas truck mass, the Baton Rouge rail +/- and the BCU must be changed in one edit; the BCU then draws 1.2507707 t more on certificate 13-47206-50000. If the corrected Opelousas truck mass is entered as its own datapoint rather than linked to the Opelousas rail Mass of load, the evidence net is 362.069096 t (recompute section 12). [log:certify.calc_view, recompute.findings, recompute.lines, policy.bcu_fix_rule]	Certify Calculation View
9.9	All datapoints have descriptive source names	Flag	'image (44).png' is a generic filename on the Opelousas rail distance; the image is a Watco email ('The total milage for this move was 2199.') naming no railcar, route or date. 'Standard Emission Factors + Calculations (8).xlsx' is uploaded as three source records. The Standard EF sheet cites 'GLEC V3.2 2025' while the attached framework PDF is 'v3.1 edition, revised and updated (2024)', and the two Baton Rouge legs cite a separate 'GLEC V3.2' library source. The four 'Aug-05/Aug-07/Aug-12/Sep-02-2026-distance-route_map.png' files show the AECN-to-Opelousas road route (3,672.1 km) yet are linked to the AECN-SOPOR truck allocation; they are four different images (SHA-256 differ). [log:certify.sources_downloaded_and_opened, certify.mangrove_map_hashes, standard_efs, certify.certify_sources_enumeration]	Certify Datapoints

Batch ID: INJ-09-2026-W4-AECN | Reporting Period: September 2026 RP, Certify period 19 Sep - 25 Sep 2026 (Ops week 2026-09-18 to 2026-09-25) | Reviewer: Claude QA Agent | QA date: 2026-10-07 | Injection Site: Basco 6 | Oil Type: AECN rail (SHPX215053 offloads 1-4 via the Baton Rouge transload; SHPX215039 and SHPX215091 offloads 1-4 and SHPX215079 offloads 3-4 via Opelousas; no direct trucks) | Certify: rmv_1M3Z5YJN51S0SN1Q | Scope: the 13 AECN components; the September site-emission components on the same removal (activities 20.550448 t, BCUs 3.10042 t) are checked by the site-emissions QA

9.10	All Sources match files in the batch folder / Ops Notes links	Fail	72 source records enumerated with their linked datapoints; the 28 scale tickets match Drive byte for byte. (1) Both Ops Notes copies in Certify are earlier versions of the sheet: src_1M3VZCDQF1 (linked to 'Bio-Oil Mass - AECN via Rail' and the rail allocated distance) shows K10 2199, L10 42.9, M42 7,598.344118 and N42 1,201.2 with no railyard split; src_1M3W1NWED1 ('- 2026-10-01', linked to the loss) shows M42 7,551.344118 and N42 1,201.2. Certify's 15,102.69 mi, 1,946 mi, 77,228.57 kg and 190,578.54 kg appear only in the live sheet, so a verifier cannot trace them from what is attached. (2) Both attached copies show SHPX215079's loss G13 as -392.76368 kg (8.2); the older copy's M13 note gives the over-draw as 'about 390.76 kg' on a raw total of '78,260.76 kg' (true 392.76368 and 78,262.76368 kg), and its M10 note 'BOL sequence has a gap, no 2499-R, not this car, unresolved.' is answered by W2's direct-truck BOL 2499-T. (3) The nine W4 BOLs, the SGS certificate, the Baton Rouge truck map and the BCU certificate are in Drive but not in Certify. (4) The attachment gate (per-batch evidence matrix built from this removal's components) fails on 11 datapoints: Mass of Injectant (per injection); Mass of load [SOPOR to Opelousas Rail Transport Process Emissions]; Mass of load [SOPOR to Baton Rouge Rail Process Emissions]; Mass of load [Baton Rouge to Basco Tanker Truck Process Emissions]; Carbon Content of Bio-oil Injected (per injection); Allocated Distance Opelousas to B6 (truck); Allocated Distance AECN to Sopor Quebec (truck); Distance traveled [SOPOR to Baton Rouge Rail Process Emissions]; Distance traveled [Baton Rouge to Basco Tanker Truck Process Emissions]; Constant CO_2 reduction [BCU - Transport Process]; AECN Bio-oil Loss - Upstream Emissions. [log:certify.ops_notes_copies, prior_weeks.w2_bol_2499T_present, certify.certify_sources_enumeration, certify.presence_checks, certify.attachment_gate, certify.required_evidence, scale_tickets.certify_vs_drive, recompute.findings]	Certify Sources
SIDEBAR - internal record (not a Certify source)				
S.1	BCU certificate capacity and usage ledger	Flag	Charm EAC Verification Inventory tracker, BCU Inventory row 13-47206-50000 (the certificate in 9.5): total reduction 2765.3612 MT, used 586.8777, remaining 2178.4834, so the 19.339 t is covered. The certificate states 2,795 TERCs at one tCO2e each, 29.6388 t more than the tracker total. The September usage row reads 171.9503 t for 9/1/2026 to 9/30/2026 with the note 'revise if Sept site emissions change', so it is provisional until the September statement exists. That row matches the statement report v1.7 (171.95 t) but not the results workbook, whose 11 September removals sum to 172.6573 t: the tracker is 0.7070 t short, and the corrected W4 BCU (9.6) would add 1.2507707 t more. [log:rp_status.results_workbook, rp_status.statement_report_v1_7, recompute.findings, bcu_sources.inventory_bcu_row, bcu_sources.certificate_terc_count, bcu_sources.usage_september, _derived_values.eac_calc]	EAC tracker